

PRELIMINARY ECOLOGICAL APPRAISAL

Preliminary Ecological Appraisal - Rossbeigh Beach Survey

Prepared For:	Client
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I. Introduction

The Rossbeigh Beach Survey, designated as Site Code RBS-2026-443, is an ecological assessment conducted to evaluate the potential impacts of proposed activities on the ecological integrity of Rossbeigh Beach and its surrounding habitats. Located at grid reference A 529 750, within County Kerry's townland, the site encompasses a diverse range of coastal habitats, including dunes and salt marshes, which are integral to the local biodiversity.

Purpose and Scope of the PEA

The primary purpose of this Preliminary Ecological Appraisal (PEA) is to provide a comprehensive understanding of the ecological features present at Rossbeigh Beach. This appraisal aims to identify potential ecological constraints and inform future management and mitigation strategies. The assessment includes a series of field surveys conducted in January 2026, covering habitats, birds, mammals, and vegetation. These surveys are essential for establishing baseline ecological conditions and identifying any protected species or habitats that may be impacted by the proposed activities.

Relevant Legislation

The ecological assessment is conducted in accordance with Irish and EU legislation, ensuring compliance with the Wildlife Acts 1976-2021, which protect native flora and fauna. Additionally, the EU Habitats and Birds Directives, transposed into Irish law through the European Communities (Birds and Natural Habitats) Regulations 2011, provide a legal framework for the conservation of natural habitats and wild species, particularly those of European importance. These directives mandate the protection of designated sites and species, necessitating thorough ecological evaluations like this.

Report Structure Overview

This report is structured to provide a detailed account of the ecological features and potential impacts associated with the Rossbeigh Beach site. It includes

- **Introduction:** Outlining the project background, site location, and scope of the assessment.
- **Methodology:** Detailing the survey methods and data collection techniques employed during the fieldwork.
- **Results:** Presenting findings from the habitat, bird, mammal, and vegetation surveys.
- **Discussion:** Interpreting the results in the context of relevant legislation and conservation objectives.
- **Recommendations:** Suggesting further surveys or mitigation measures to address identified ecological constraints.

Given the proximity of the Castlemaine Harbour SAC, with its diverse estuarine and coastal habitats, this PEA is crucial for ensuring that any proposed developments do not adversely affect the site's ecological integrity. The report highlights the need for continued monitoring and potential additional surveys to fully characterise the site's ecological value and inform sustainable management practices.

II. Methodology

Desk Study

The desk study for the Rossbeigh Beach Survey involved a comprehensive review of existing ecological data from several key sources. The National Parks and Wildlife Service (NPWS) database was consulted, revealing the proximity of the Castlemaine Harbour SAC (Site Code: 000343) directly adjacent to the project site. The Global Biodiversity Information Facility (GBIF) and the National Biodiversity Data Centre (NBDC) provided records of protected species, including the Meadow Pipit (*Anthus pratensis*), located 5.98 km from the site. Water quality data were obtained from

m the Environmental Protection Agency (EPA) for nearby water bodies, specifically Nambrack darrig Glenbeigh and Coomnacronia, both within the Laune-Maine-Dingle Bay catchment. Catchm ents.ie was also reviewed to understand hydrological connectivity and potential impacts on

Field Survey

Four field surveys were conducted as part of this assessment:

- **Habitat Survey:** Conducted on 15th January 2026 from 09:00 to 16:30. The weather was suitable, with moderate winds and 60% cloud cover. The survey covered the main beach, embryo dunes, marra m dunes, fixed dune grassland, and dune slack area
- **Bird Survey:** Carried out on 16th January 2026 from 06:30 to 11:00. Conditions were favourable for bird observations with light winds and 40% cloud cover. The survey focused on wintering birds along th e shoreline and dune system, coinciding with a receding tid
- **Mammal Survey:** Conducted on 17th January 2026 from 08:00 to 13:00. Despite light rain early in the day, conditions remained conducive for detecting mammal signs. The survey targeted otter activity an d badger sett
- **Vegetation Survey:** Performed concurrently with the habitat survey on 15th January 2026. The survey recorded DAFOR abundance for key species across various habitats

Habitat Mapping

Habitat mapping followed the Fossitt (2000) classification system, identifying seven distinct habitat types across the 18.10-hectare survey area. The mapping was conducted using standard techniques , including GPS for accurate boundary delineation and detailed field notes to describe habit at conditions and threa

Limitations

Several limitations were identified during the assessment:

- **Seasonal Constraints:** Surveys were conducted in January, which may not capture the full range of seasonal species presence, particularly for flora and breeding birds. Further surveys in spring an d summer are recommended to address these gap
- **Weather Conditions:** While generally suitable, light rain during the mammal survey may have affected the detection of some signs
- **Access Restrictions:** Certain areas of the dune system were difficult to access due to natural barriers, potentially limiting the comprehensiveness of the survey in those regions
- **Survey Effort:** The single winter bird survey is insufficient for a complete characterisation of overwintering bird assemblages. Additional surveys are recommended to establish peak numbers an d usage pattern

These limitations highlight the need for continued monitoring and additional surveys to ensure a comprehensive ecological understanding of the site, in line with CIEEM and NRA guidelines

III. Results - Designated Sites

Designated Sites Summary

The following table summarises the designated sites within proximity to the Rossbeigh Beach Survey project area

Designated Site	Site Code	Designation	Distance (km)	Notes
Castlemaine Harbou	000343	SAC	0.00	Estuaries, Mudflat

Castlemaine Harbour SAC (Site Code: 000343)

Castlemaine Harbour SAC is directly adjacent to the project site, indicating potential direct connectivity and impact pathways. The site encompasses a variety of habitats of community interest , including estuarine and coastal habitats such as estuaries, mudflats, and vegetated sea cliffs. T

he conservation status of these habitats is predominantly inadequate or bad, with only the vegetated sea cliffs currently assessed as favourable. The site is under significant pressure from pollution, development, aquaculture, and recreational activities. Threats to the site include eutrophication, climate change, invasive species, and sea level rise. Given the proximity, any proposed activities at Rossbeigh Beach should be carefully assessed for potential impacts on these sensitive habitats.

Potential Connectivity and Impact Assessment

The direct adjacency of Castlemaine Harbour SAC to the project site suggests potential connectivity, particularly concerning hydrological and ecological linkages. The dune systems and coastal habitats at Rossbeigh Beach may serve as transitional zones or corridors for species movement between the project site and the SAC. The presence of protected species such as the otter (*Lutra lutra*) and various bird species, including the Meadow Pipit (*Anthus pratensis*), further underscores the ecological connectivity between these areas.

Given the potential for direct and indirect impacts, an Appropriate Assessment (AA) Screening is recommended to determine whether the project could significantly affect the integrity of the Castlemaine Harbour SAC. This assessment should consider the cumulative pressures and threats identified, including recreational pressure, erosion, and potential pollution pathways.

Data Gaps and Recommendations

- **Appropriate Assessment (AA) Screening:** Conduct an AA Screening to evaluate potential impacts on Castlemaine Harbour SAC, focusing on hydrological connectivity and species movement corridors.
- **Additional Surveys:** Given the limited number of surveys conducted, further ecological surveys are recommended to fully characterise the site's biodiversity and assess potential impacts. This includes additional winter bird surveys to capture peak numbers and bat surveys during the summer months.
- **Monitoring and Mitigation:** Implement monitoring programmes to track changes in habitat conditions and species populations. Develop mitigation measures to address identified threats, such as recreational pressure and invasive species management.

Caveats

The conclusions drawn in this section are based on the available data and desk research findings. The absence of comprehensive survey data for certain taxa and habitats may limit the accuracy of impact predictions. Further detailed studies are necessary to ensure a robust ecological appraisal and to inform any necessary mitigation strategies.

III. Results - Habitats

The habitat survey conducted at Rossbeigh Beach identified a total of seven distinct habitat types, classified according to Fossitt (2000). These habitats cover an area of 18.10 hectares and include a range of dune systems, grasslands, and marshes, each with varying conditions and ecological significance.

CD2 — Marram Dunes (3.20 ha)

- **Condition:** Good
- **EU Annex I:** 2120
- **Evaluation:** National
- **Notable Features:** The marram dunes are well-developed along the main beach frontage, with *Ammophila arenaria* (Marram Grass) as the dominant species. The seaward face is characterised by *Elymus farctus*. Despite some blow-out areas, these are actively re-colonising, indicating a dynamic and resilient system.
- **Threats:** Recreational pressure, erosion, and climate change, particularly sea-level rise, pose significant threats to this habitat.

CD3 — Fixed Dunes and Dune Grassland (5.80 ha)

- **Condition:** Good
- **EU Annex I:** 2130
- **Evaluation:** International
- **Notable Features:** This species-rich habitat is located behind the mobile dune ridge and supports a diverse flora, including *Festuca rubra* and *Galium verum*. Noteworthy is the presence of orchid species such as *Dactylorhiza fuchsii*, highlighting the habitat's ecological value
- **Threats:** Undergrazing, scrub encroachment, and invasive species are identified as primary threats, potentially altering the habitat's structure and species composition

CD5 — Dune Slacks (1.40 ha)

- **Condition:** Moderate
- **EU Annex I:** 2190
- **Evaluation:** International
- **Notable Features:** The dune slacks are seasonally wet areas situated between the main dune ridge and fixed grassland. *Salix repens* is abundant, with *Hydrocotyle vulgaris* and *Mentha aquatica* also present. The water table is within 0.5 metres of the surface, supporting a unique assemblage of species
- **Threats:** Drainage, scrub encroachment, and eutrophication threaten the hydrological and ecological integrity of this habitat

CD1 — Embryo Dunes (0.80 ha)

- **Condition:** Good
- **EU Annex I:** 2110
- **Evaluation:** National
- **Notable Features:** These narrow bands of dunes are located along the upper strandline, dominated by *Elymus farctus*, with *Cakile maritima* and *Salsola kali* also present. Active sand accumulation is a positive indicator of habitat health
- **Threats:** Trampling, storm erosion, and beach cleaning activities are significant threats, potentially leading to habitat degradation

CM1 — Upper Salt Marsh (2.10 ha)

- **Condition:** Good
- **EU Annex I:** 1330
- **Evaluation:** National
- **Notable Features:** Located on the sheltered eastern side of the spit, this habitat is co-dominated by *Festuca rubra*, *Armeria maritima*, and *Plantago maritima*. The well-developed creek system enhances its ecological complexity
- **Threats:** Coastal squeeze and agricultural runoff are primary concerns, potentially impacting the habitat's ecological functions

GA1 — Improved Agricultural Grassland (4.50 ha)

- **Condition:** Poor
- **Evaluation:** Local
- **Notable Features:** This intensively managed grassland is adjacent to the dune system, dominated by *Lolium perenne* and *Trifolium repens*. The habitat exhibits low species diversity, reflecting its agricultural use
- **Threats:** Intensification and fertiliser application are ongoing threats, reducing ecological value and biodiversity

WL1 — Hedgerows (0.30 ha)

- **Condition:** Moderate
- **Evaluation:** High Local
- **Notable Features:** The hedgerows, primarily composed of *Crataegus monogyna* and *Prunus spinosa*, provide essential connectivity between the dune grassland and the wider countryside, serving a

s ecological corridors for various specie

- **Threats:** Gap creation and agricultural intensification threaten the structural integrity and ecological function of these hedgerows

Habitat Connectivity and Ecological Corridors The Rossbeigh Beach site demonstrates significant habitat connectivity, particularly through the hedgerows and dune systems, which serve as ecological corridors facilitating species movement and genetic exchange. The proximity to Castlemaine Harbour SAC further underscores the importance of maintaining these connections to support regional biodiversity

Threats and Evaluation Notes The primary threats across the habitats include recreational pressure, climate change impacts, invasive species, and agricultural practices. The presence of EU Habitats Directive Annex I types within the site highlights its ecological importance, necessitating careful management to mitigate identified threats. Further surveys, particularly during breeding seasons and additional bird counts, are recommended to fully assess the site's ecological dynamics and inform conservation strategies

III. Results - Fauna

The fauna surveys conducted at Rossbeigh Beach (Site Code: RBS-2026-443) recorded a diverse assemblage of species across several taxonomic groups, with particular emphasis on birds and mammals. The surveys were carried out under suitable weather conditions, ensuring reliable data collection

Bird Observations:

A total of ten bird species were recorded, many of which are protected under the Wildlife Acts 1976-2021 and the EU Birds Directive. Notable observations include

- **Charadrius hiaticula** (Ringed Plover): Twelve individuals observed feeding on the strandline. This species is protected and was observed in winter plumage, indicating its use of the site as a wintering ground
- **Calidris alba** (Sanderling): Twenty-eight individuals were actively feeding along the tide edge, displaying characteristic wave-chasing behaviour
- **Haematopus ostralegus** (Oystercatcher): A flock of forty-five was noted on exposed sandflats, with some individuals roosting on the upper beach
- **Numenius arquata** (Curlew): Eight individuals were recorded feeding at the salt marsh edge. The Curlew is a red-listed species under the Birds of Conservation Concern in Ireland (BoCCI), highlighting its conservation importance
- **Pluvialis apricaria** (Golden Plover): A significant flock of 120 was observed in the fixed dune grassland, indicating the site's importance for this species during winter
- **Alauda arvensis** (Skylark): Three individuals were noted in song flight over the fixed dune grassland. This species is amber-listed under BoCCI
- **Falco peregrinus** (Peregrine Falcon): A single adult was observed hunting over the dune system, demonstrating the site's role as a hunting ground
- **Linaria cannabina** (Linnet): Fifteen individuals were feeding on seed heads in the dune grassland. This species is also amber-listed under BoCCI
- **Tadorna tadorna** (Shelduck): Six individuals were observed in pairs on the salt marsh creek, suggesting potential breeding activity in the area
- **Anthus pratensis** (Meadow Pipit): Eight individuals were scattered across the fixed dune grassland and salt marsh edge, consistent with desk study records indicating its presence within the vicinity

Mammal Observations:

Five mammal species were recorded, with several protected under the Wildlife Acts:

- **Lutra lutra** (Otter): Signs of otter activity, including fresh spraint and prints, were found at the stream mouth. This indicates regular use of the area, though no holt was identified

- **Lepus timidus hibernicus** (Irish Hare): Two individuals were observed feeding in the fixed dune grassland, underscoring the habitat's suitability for this protected species
- **Oryctolagus cuniculus** (Rabbit): Extensive burrow systems were noted in the fixed dune grassland, contributing to habitat structure and maintenance
- **Meles meles** (Badger): An active two-entrance sett was found at the edge of the dune grassland, with fresh bedding material indicating recent use
- **Erinaceus europaeus** (Hedgehog): Droppings were found along the hedgerow, suggesting presence, though no individuals were observed

Ecological Importance and Recommendations:

The assemblage of bird and mammal species observed at Rossbeigh Beach highlights the site's ecological significance, particularly for wintering and breeding birds. The presence of protected species such as the Curlew and Irish Hare further underscores the need for careful management and conservation effort

- **Further Surveys:** Additional winter bird surveys are recommended to fully characterise the overwintering bird assemblage. A minimum of three additional counts between November and March is advised
- **Otter Monitoring:** Continued monitoring of otter activity is recommended, particularly during the breeding season, to confirm the presence of holts or couches
- **Badger Protection:** Any works within 50 metres of the identified badger sett will require a derogation licence from NPWS, with a 30-metre exclusion zone applicable during the breeding season (December to June)

Overall, the data collected provides a robust foundation for assessing the ecological value of Rossbeigh Beach and informing future management and conservation strategies. However, the limitations of a single winter bird survey should be addressed to ensure comprehensive understanding and protection of the site's biodiversity

III. Results - Flora

The flora survey conducted at Rossbeigh Beach identified a diverse assemblage of plant species across the various dune habitats, with notable occurrences of both native and non-native species. The survey adhered to the Fossitt (2000) habitat classification, covering embryo dunes (CD1), marram dunes (CD2), fixed dunes and dune grassland (CD3), and dune slacks (CD5), among others

Key Species Observations:

- **Ammophila arenaria (Marram Grass)** was recorded as dominant (DAFOR: D) within the marram dune habitat (CD2), playing a crucial role in dune stabilisation through its extensive rhizome system
- **Dactylorhiza fuchsii (Common Spotted-orchid)** was observed as occasional (DAFOR: O) in the fixed dune grassland (CD3), with approximately 30 flowering spikes noted, indicating a healthy orchid population
- **Anacamptis pyramidalis (Pyramidal Orchid)** was rare (DAFOR: R) in the same habitat, with only five individuals located, highlighting the calcareous nature of the dune grassland
- **Hippophae rhamnoides (Sea-buckthorn)**, an invasive non-native species, was noted as occasional (DAFOR: O) at the edges of the fixed dune grassland, necessitating management to prevent further spread
- **Salix repens (Creeping Willow)** was abundant (DAFOR: A) in the dune slack habitat (CD5), forming a low mat with *Hydrocotyle vulgaris*
- **Eryngium maritimum (Sea Holly)**, a Flora Protection Order species, was rare (DAFOR: R) on the embryo dune/mobile dune interface, with eight plants recorded. This species is vulnerable to trampling and disturbance, warranting protective measures

- **Armeria maritima (Thrift)** was frequent (DAFOR: F) on the upper salt marsh (CM1), forming characteristic cushions

Flora Protection Order Species:

- **Eryngium maritimum (Sea Holly)** is protected under the Flora Protection Order 2022. Its presence on the site requires careful management to mitigate impacts from recreational activities

Data Gaps and Recommendations:

- The presence of invasive species such as **Hippophae rhamnoides** necessitates a management plan to control its spread, particularly in the fixed dune grassland
- Further monitoring of **Eryngium maritimum** populations is recommended to ensure their protection and to assess the effectiveness of any mitigation measures implemented

The survey results underscore the ecological value of the Rossbeigh Beach dune system, with its diverse plant communities contributing to the site's overall biodiversity. Continued monitoring and management are essential to maintain the ecological integrity of these habitats, particularly in light of potential threats such as recreational pressure and invasive species encroachment.

III. Results - Invasive Species

During the field surveys conducted at Rossbeigh Beach, one invasive species, **Hippophae rhamnoides** (Sea-buckthorn), was identified within the fixed dune grassland (CD3). This species is listed under the Third Schedule of the European Communities (Birds and Natural Habitats) Regulations 2011, which highlights its potential to cause ecological harm. The Sea-buckthorn is currently colonising approximately 0.2 hectares at the southern boundary of the fixed dune grassland, spreading from previously planted specimens. Its presence poses a threat to the native biodiversity and the integrity of the Annex I habitat 2130 (Fixed dunes with herbaceous vegetation).

No other invasive species were observed during the surveys or recorded in the desk study. However, the habitats present, particularly the fixed dunes and dune grassland (CD3), are susceptible to colonisation by invasive species due to existing pressures such as undergrazing and scrub encroachment. The potential for further invasive species colonisation should be monitored, especially in areas where recreational pressure and habitat disturbance are evident.

Recommendations:

- **Biosecurity Measures:** Implement biosecurity protocols to prevent the introduction and spread of invasive species. This includes cleaning equipment and footwear before and after site visits and ensuring that any materials brought onto the site are free from invasive species.
- **Management Plan:** Develop and implement a management plan for the control of Sea-buckthorn. Mechanical removal during the winter months is advised to minimise disturbance to native flora and fauna.
- **Monitoring:** Regular monitoring of the site for invasive species should be conducted, particularly in the fixed dune grassland and other vulnerable habitats, to ensure early detection and rapid response to any new invasion.

These measures will help maintain the ecological integrity of the habitats at Rossbeigh Beach and support compliance with relevant Irish and EU wildlife legislation.

IV. Evaluation

The ecological evaluation of the Rossbeigh Beach site, as part of the Rossbeigh Beach Survey (Site Code: RBS-2026-443), considers several key habitats and species, assessing their ecological value based on criteria such as naturalness, rarity, size/extent, diversity, fragility, typicalness, recorded history, position in ecological unit, and potential value. The evaluation also applies a geographic frame of reference for each receptor, ranging from local to international significance.

Habitats:

- **Marram Dunes (CD2):**
- **Extent & Condition:** 3.20 ha, good condition.
- **Value:** National significance, EU Annex I habitat (2120).
- **Threats:** Recreational pressure, erosion, climate change.
- **Evaluation:** The well-developed marram dune system is typical of coastal dune habitats and plays a crucial role in stabilising the beach frontage. Its naturalness and resilience to erosion highlight its ecological importance.
- **Fixed Dunes and Dune Grassland (CD3):**
- **Extent & Condition:** 5.80 ha, good condition.
- **Value:** International significance, EU Annex I habitat (2130).
- **Threats:** Undergrazing, scrub encroachment, invasive species.
- **Evaluation:** This species-rich habitat supports notable orchid species and is a prime example of a diverse dune grassland ecosystem, contributing significantly to the site's biodiversity.
- **Dune Slacks (CD5):**
- **Extent & Condition:** 1.40 ha, moderate condition.
- **Value:** International significance, EU Annex I habitat (2190).
- **Threats:** Drainage, scrub encroachment, eutrophication.
- **Evaluation:** The seasonally wet dune slacks are fragile and sensitive to hydrological changes, underscoring their ecological importance within the dune system.
- **Embryo Dunes (CD1):**
- **Extent & Condition:** 0.80 ha, good condition.
- **Value:** National significance, EU Annex I habitat (2110).
- **Threats:** Trampling, storm erosion, beach cleaning.
- **Evaluation:** These dunes are critical for sand accumulation and coastal dynamics, with their naturalness and fragility making them a key ecological receptor.
- **Upper Salt Marsh (CM1):**
- **Extent & Condition:** 2.10 ha, good condition.
- **Value:** National significance, EU Annex I habitat (1330).
- **Threats:** Coastal squeeze, agricultural runoff.
- **Evaluation:** The salt marsh supports a well-developed creek system and is vital for flood regulation and habitat diversity.
- **Improved Agricultural Grassland (GA1):**
- **Extent & Condition:** 4.50 ha, poor condition.
- **Value:** Local significance.
- **Threats:** Intensification, fertiliser application.
- **Evaluation:** This habitat has low species diversity and limited ecological value, primarily serving as a buffer to more sensitive areas.
- **Hedgerows (WL1):**
- **Extent & Condition:** 0.30 ha, moderate condition.
- **Value:** High local significance.
- **Threats:** Gap creation, agricultural intensification.

- **Evaluation:** The hedgerows provide important connectivity and habitat for fauna, enhancing landscape biodiversity

Species:

- **Birds:** The site supports a range of protected bird species, including the Red-listed Curlew (*Numenius arquata*) and Amber-listed Skylark (*Alauda arvensis*). The presence of these species, along with others like the Peregrine Falcon (*Falco peregrinus*), highlights the site's importance for avian biodiversity at a national level.
- **Mammals:** The presence of protected species such as the Otter (*Lutra lutra*) and Irish Hare (*Lepus timidus hibernicus*) indicates the site's value for mammalian biodiversity. The otter's use of the stream mouth and the presence of a badger sett further underscore the site's ecological significance.
- **Plants:** The occurrence of protected and rare plant species, including the Flora Protection Order species Sea Holly (*Eryngium maritimum*), adds to the site's botanical value, particularly within the dune habitats.

Designated Sites:

- **Castlemaine Harbour SAC:** The proximity of the site to Castlemaine Harbour SAC, with its diverse range of habitats and species, necessitates careful consideration of potential impacts, particularly given the SAC's inadequate conservation status for several habitats.

Recommendations:

- Further surveys are recommended to fully characterise the winter bird assemblage and assess potential impacts on the Meadow Pipit population.
- A summer bat survey should be conducted to evaluate potential impacts on foraging habitats.
- Management plans should address invasive species control, particularly the spread of Sea-buckthorn.

The evaluation highlights the site's significant ecological value, particularly its dune and salt marsh habitats, and the need for targeted conservation measures to mitigate identified threats.

V. Discussion

The ecological appraisal of Rossbeigh Beach has identified several key habitats and species of conservation concern, necessitating careful consideration of potential impacts arising from any proposed developments. The site is characterised by a diverse dune system, including **marram dunes (CD2)**, **fixed dunes and dune grassland (CD3)**, **dune slacks (CD5)**, and **embryo dunes (CD1)**, each with varying conservation evaluations from national to international importance. Additionally, the proximity of the **Castlemaine Harbour SAC** underscores the necessity for a thorough impact assessment, particularly given the site's inadequate conservation status and the pressures.

Potential Impacts on Ecological Receptors:

- **Habitat Degradation:** The dune habitats, particularly the **fixed dunes (CD3)** and **dune slacks (CD5)**, are vulnerable to threats such as undergrazing, scrub encroachment, and drainage. The presence of invasive species like **Hippophae rhamnoides** further exacerbates these threats, potentially leading to habitat degradation. Management strategies, including mechanical removal of invasive species, are recommended to mitigate these impacts.
- **Species Disturbance:** The site supports a range of protected bird species, including **Golden Plover** and **Curlew**, both of which are sensitive to disturbance. Recreational activities, particularly dog walking, have already been observed to cause disturbance to these species. Implementing measures such as designated pathways and information signage could help minimise human-induced disturbance.
- **Protected Flora:** The presence of **Sea Holly (*Eryngium maritimum*)**, a species protected under the Flora Protection Order 2022, highlights the need for protective measures against trampling and disturbance, particularly in the embryo dunes. Fencing and boardwalks could be effective in preserving the

se sensitive are

Cumulative and In-Combination Effects:

The cumulative impacts of recreational pressure, climate change, and invasive species could significantly alter the ecological character of the site. In combination with the existing pressures on the Castlemaine Harbour SAC, there is a risk of exacerbating habitat degradation and species disturbance. The precautionary principle should be applied, particularly where uncertainties exist regarding the full extent of these cumulative impacts.

Professional Opinion and Recommendations:

The ecologist's professional opinion underscores the need for further surveys to fill existing data gaps. Specifically, additional winter bird surveys are recommended to better characterise the overwintering bird assemblage, and a bat survey is advised during the summer months to assess potential impacts on foraging habitats. Furthermore, the presence of **otter spraint** at the stream mouth suggests regular use of the area by otters, necessitating further investigation during the breeding season to ensure compliance with the Wildlife Acts 1976-

Triggers for Appropriate Assessment Screening:

Given the direct connectivity to the Castlemaine Harbour SAC and the presence of habitats and species of community interest, an Appropriate Assessment Screening is warranted. This will determine whether the proposed activities are likely to have significant effects on the SAC, either alone or in combination with other plans or projects.

In conclusion, the Rossbeigh Beach site presents a complex ecological landscape with significant conservation value. Any proposed developments must be carefully assessed to avoid adverse impacts on the site's habitats and species, with particular attention to the cumulative effects and the need for further detailed survey.

VI. Recommendations

Further Surveys

- **Winter Bird Surveys:** Conduct a minimum of three additional WeBS-type counts between November and March to establish peak numbers and full characterisation of overwintering bird assemblages
- **Bat Surveys:** Undertake walked transect surveys with bat detectors during May to September to assess potential bat foraging activity along hedgerows and dune slacks
- **Otter Surveys:** Conduct targeted surveys during the breeding season (April to June) to confirm otter activity and potential holt locations along the beach and dune interface

Mitigation Measures

- **Avoidance:** Implement a 30-metre exclusion zone around the identified badger sett from December to June to avoid disturbance during the breeding season, as required under the Wildlife Acts
- **Minimisation:** Erect temporary fencing and informational signage to guide pedestrian traffic and minimise trampling on embryo dunes and Sea Holly colonies
- **Remediation:** Develop a management plan for the mechanical removal of invasive Sea-buckthorn during winter months to prevent further encroachment into fixed dune grassland
- **Compensation:** If habitat loss is unavoidable, create compensatory habitats such as additional dune slack areas to offset impacts on Annex I habitats

Enhancement Opportunities

- **Biodiversity Net Gain:** Establish native plantings along hedgerows to enhance connectivity and provide additional foraging resources for birds and bats. Consider planting native dune species to bolster existing habitat
- **Habitat Restoration:** Restore degraded areas of dune slack by reintroducing native flora and managing

water levels to support diverse wetland communities

Monitoring

- **Ongoing Monitoring:** Implement a long-term monitoring programme for key habitats and species, particularly focusing on the condition of Annex I habitats and the presence of protected species such as otters and badger
- **Invasive Species Control:** Regularly monitor and manage invasive species, particularly Sea-buckthorn, to prevent further spread and impact on native biodiversity

Timing Constraints

- **Seasonal Restrictions:** Avoid works within 50 metres of the badger sett from December to June, unless a derogation licence is obtained from NPWS. Schedule invasive species removal and habitat restoration activities outside of sensitive periods for breeding birds and mammal
- **Optimal Survey Periods:** Align further surveys with optimal periods for target species, ensuring accurate and comprehensive data collection to inform subsequent ecological assessments and management plan

Caveats

- **Data Gaps:** Current data is limited to winter surveys; additional seasonal data is required to fully assess the ecological value and potential impacts on the site. Further surveys as recommended are essential to address these gaps and inform effective conservation strategies

VII. Appendices

Appendix A: Site Location Map

- This appendix contains a detailed map illustrating the location of the Rossbeigh Beach Survey site, identified by Site Code RBS-2026-443. The map includes the grid reference A 529 750, providing spatial context for the survey area

Appendix B: Habitat Map

- This appendix presents a habitat map of the survey area, with habitats classified according to Fossitt (2000) codes. The map delineates the seven identified habitat types, including CD2 (Marram dunes), CD3 (Fixed dunes and dune grassland), and others, along with their respective areas and condition

Appendix C: Species List

- A comprehensive list of species observed during the field surveys is included in this appendix. It details observation counts and evidence types for each species, highlighting protected species and their conservation status under relevant Irish and EU legislation

Appendix D: Site Photographs

- This appendix features a collection of photographs taken during the field surveys, providing visual documentation of the habitats, species, and key features of the Rossbeigh Beach site

Appendix E: Survey Datasheets

- Survey datasheets for each of the four completed surveys are compiled in this appendix. These datasheets include detailed notes on survey conditions, methodologies, and specific observations recorded during the habitat, bird, mammal, and vegetation survey

Appendices

Appendix A: Habitat Map

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